

[Midterm Exam - (SOLUTION)]

Spring 2025: CSE 309: Theory of Automata

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Thursday, March 20, 2025. Total Marks: 60, Duration: 120 minutes

- **PLEASE READ THESE INSTRUCTIONS CAREFULLY.**

- Read all questions carefully before you start answering and prioritize which questions to attempt first.
- Please write your name and student ID on this question paper clearly.
- Please do not use pencils or red/green/yellow pens for answers that you want to be graded.
- No queries during the exam.
- If we can't read we can't grade.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	0	
Total:	60	

– Good Luck –

Question 1

[10 points]

Mark **T** for True and **F** for False.

- (a) **T** The regular expressions $(a \cup b)^*$ and $(b^*a^*)^*$ generate the same language.
- (b) **T** If A and B are *regular languages* then $A \setminus B$ (defined as $A \setminus B = \{w : w \in A \text{ and } w \notin B\}$) is also a *regular language*.
- (c) **F** If A and B are two *context-free languages* then $A \cap B$ is also a *context-free language*.
- (d) **T** If L is a *regular language* then L^2 is also a *regular language*.
- (e) **F** If L is a Turing-decidable language then its complement $\bar{L}$ is not Turing-decidable.
- (f) **T** If $L_1, L_2, \dots, L_n$ are n different Turing-decidable languages over some alphabet Σ then $L_1 \cap L_2 \cap \dots \cap L_n$ is also Turing-decidable.
- (g) **F** Two languages A and B are equal if and only if $A \cap \bar{B} = \emptyset$.
- (h) **F** Every subset of a context-free language is context-free.
- (i) **F** Let G is given in Chomsky Normal Form. Then any string $w \in L(G)$ has at most $|w|$ steps in its left-most derivation.
- (j) **F** The tape alphabet Γ can be the same as the input alphabet Σ .

Question 2

[10 points]

Let $\Sigma = \{0, 1\}$, and let A be the language of strings $w \in \Sigma^*$ containing exactly one double symbol. (A string has a double symbol if it contains 00 or 11 as a substring. The string 10010 has exactly one double symbol, but 100010 has two double symbols.) Give a regular expression for A and describe a DFA that recognizes A .

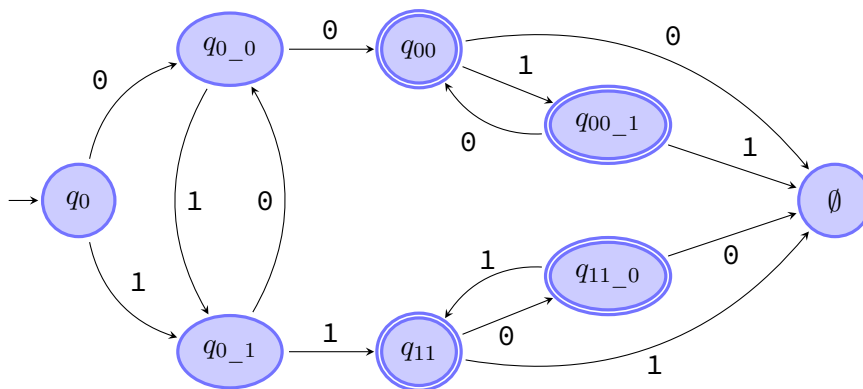
Solution: Regular expression for A :

$$(X \cdot 00 \cdot X) \cup (X \cdot 11 \cdot X)$$

where

$$X = (01^*(\epsilon \cup 0)) \cup (10^*(\epsilon \cup 1)).$$

Following DFA M accepts A :



Question 3

[10 points]

Use the *pumping lemma* to show that the following language L is not regular:

$$L = \{\mathbf{a}^k w \mid w \in \{\mathbf{a}, \mathbf{b}\}^*, |w| = k, k \geq 0\}.$$

Recall that the pumping lemma for regular languages states that for every alphabet Σ and every regular language $A \subseteq \Sigma^*$, there exists a pumping length $p \geq 1$ for A , such that for every string $w \in A$ with $|w| \geq p$, there exist strings $x, y, z \in \Sigma^*$, with $w = xyz$, such that

1. $|y| \geq 1$
2. $|xy| \leq p$, and
3. $xy^iz \in A$, for all $i \geq 0$.

Solution: Assume for the sake of contradiction that L is regular. Let p be the pumping length given by the pumping lemma. Consider the string $s = \mathbf{a}^p \mathbf{b}^p \in L$ where $|s| = 2p \geq p$. By the pumping lemma, we can write $s = xyz$ such that $|xy| \leq p$ and $|y| \geq 1$. Since $|xy| \leq p$, y consists of only $\mathbf{a}$'s. Pumping y down will reduce the number of $\mathbf{a}$'s in the string, and the resulting string cannot be written as $\mathbf{a}^k \mathbf{b}^k$ for any k and thus not in L . This contradicts the pumping lemma, so L is not regular.

Note: Pumping up will not work (in most of the cases for all even $i \geq 2$) because half of extra $\mathbf{a}$'s can be considered part of string w .

Question 4

[10 points]

Consider the following CFG $G = (V, T, S, P)$, where $V = \{S, X, Y, B, C\}$ and $T = \{a, b, c\}$.

$$\begin{aligned} S &\rightarrow X \mid YC \\ X &\rightarrow aXc \mid B \\ Y &\rightarrow aYb \mid \epsilon \\ B &\rightarrow bB \mid \epsilon \\ C &\rightarrow cC \mid \epsilon \end{aligned}$$

- (a) (4 marks) Which of the following statement(s) about the Language $L(G)$ is(are) correct?
- i. **T** $aabbcc \in L(G)$
 - ii. **F** $\{a^i b^j c^k : i, j, k \geq 0\} \subseteq L(G)$
 - iii. **T** $\{a^i b^i c^k : i, k \geq 0\} \subseteq L(G)$
 - iv. **T** $\{a^i b^i c^i : i \geq 0\} \subseteq L(G)$
- (b) (4 marks) What is the Language generated by G ?

Solution:

$$L(G) = \{a^i b^j c^k : i, j, k \geq 0, i = j \text{ or } i = k\}$$

- (c) (2 marks) Is the above grammar ambiguous?

Solution: Yes, the above grammar is ambiguous. The string $aabbcc$ can be derived in two different ways:

1. $S \Rightarrow X \Rightarrow aXc \Rightarrow aaXcc \Rightarrow aaBcc \Rightarrow aabBcc \Rightarrow aabbBcc \Rightarrow aabbcc$
2. $S \Rightarrow YC \Rightarrow aYbC \Rightarrow aaYbbC \Rightarrow aabbccC \Rightarrow aabbccC \Rightarrow aabbcc$

Therefore, the grammar is ambiguous.

Question 5

[10 points]

Consider the Turing machine $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$ where $\Sigma = \{0, 1\}$, $\Gamma = \{0, 1, X, Y, \sqcup\}$, and the transition function δ is defined as follows:

	0	1	X	Y	$\sqcup$
q_0	(q_1, X, R)	$(q_{\text{reject}}, \sqcup, R)$	$(q_{\text{reject}}, \sqcup, R)$	(q_3, Y, R)	$(q_{\text{reject}}, \sqcup, R)$
q_1	$(q_1, 0, R)$	(q_2, Y, L)	$(q_{\text{reject}}, \sqcup, R)$	(q_1, Y, R)	$(q_{\text{reject}}, \sqcup, R)$
q_2	$(q_2, 0, L)$	$(q_{\text{reject}}, \sqcup, R)$	(q_0, X, R)	(q_2, Y, L)	$(q_{\text{reject}}, \sqcup, R)$
q_3	$(q_{\text{reject}}, \sqcup, R)$	$(q_{\text{reject}}, \sqcup, R)$	$(q_{\text{reject}}, \sqcup, R)$	(q_3, Y, R)	$(q_{\text{accept}}, \sqcup, R)$
q_{accept}	-	-	-	-	-
q_{reject}	-	-	-	-	-

Give the sequence of configurations that M enters when started on the input string 0011.

Solution:

- | | | |
|---------------|----------------|--|
| 1. $q_0 0011$ | 6. $Xq_0 0Y1$ | 11. $XXq_0 YY$ |
| 2. $Xq_1 011$ | 7. $XXq_1 Y1$ | 12. $XXYq_3 Y$ |
| 3. $X0q_1 11$ | 8. $XXYq_1 1$ | 13. $XXYYq_3 \sqcup$ |
| 4. $Xq_2 0Y1$ | 9. $XXq_2 YY$ | 14. $XXYY \sqcup q_{\text{accept}} \sqcup$ |
| 5. $q_2 X0Y1$ | 10. $Xq_2 XYY$ | |

Question 6

[10 points]

Given a string w over the alphabet $\Sigma = \{0, 1\}$, give implementation-level details of a Turing machine that decrements the binary number represented by w by 1. For example, on input $\boxed{101}$, the tape should be modified to $\boxed{100}$. Make sure that your Turing machine replaces any leading 0 with blank symbol $\sqcup$.

Solution: Let L be the language and $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$ be a Turing machine. Here, $\Sigma = \{0, 1\}$ and $\Gamma = \{0, 1, \sqcup\}$. $Q = \{q_1, q_2, q_3, q_4, q_{\text{accept}}, q_{\text{reject}}\}$ is the set of states. The transition function δ is defined as follows:

	0	1	$\sqcup$
q_1	$(q_1, 0, R)$	$(q_1, 1, R)$	$(q_2, \sqcup, L)$
q_2	$(q_2, 1, L)$	$(q_3, 0, L)$	$(q_{\text{accept}}, \sqcup, R)$
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	$(q_4, \sqcup, R)$
q_4	$(q_4, \sqcup, R)$	$(q_{\text{accept}}, 1, R)$	$(q_{\text{accept}}, \sqcup, R)$

Bonus question

[10 bonus points]

A *Turing machine with left reset* is similar to standard Turing machine, but the transition function has the form

$$\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{\mathbf{R}, \mathbf{RESET}\}.$$

If $\delta(q, \mathbf{a}) = (r, \mathbf{b}, \mathbf{RESET})$, when the machine is in state q reading an $\mathbf{a}$, the machine's head jumps to the left-hand end of the tape after it writes $\mathbf{b}$ on the tape and enters state r . Note that these machines do not have the usual ability to move the head one symbol left. Show that Turing machines with left reset recognize the class of Turing-recognizable languages.

Solution: We will show that a Turing machine with left reset can simulate a standard Turing machine.

Right transitions are the same. To move one position to the left, execute the following algorithm:

1. Mark the current position as special.
2. Move to the origin (**RESET**).
3. Translate the tape one cell to the right, while keeping the special marker in place.
4. Move to the origin (**RESET**).
5. Scan until reaching the special marker.
6. Remove the special marker.

After this algorithm, the head is one cell to the left of the original position.